SukariSol

One connected digital intelligence platform
for Kenya’s sugar value chain

MAP • CONNECT • ANALYSE • RECONCILE

Proposed by: Charles Odolo (odolocharles@gmail.com) | Emmanuel Wonka (kiprotichemmanuel360@gmail.com)

Themes: Climate-smart agronomy; regional competitiveness; financing and investment; out-grower institutions; consumer protection and market diversification; AI and industrial automation.

Abstract

Kenya’s sugar value chain connects farmers, out-grower institutions, transporters, mills, distributors and consumers, but fragmented records can limit shared visibility and timely decisions. SukariSol proposes an integrated digital platform combining geospatial and field intelligence, logistics, delivery/payment reconciliation, factory reliability, traceability, by-product opportunities and market planning. Two shared capabilities strengthen the platform: SukariSol Intelligence, which applies fit-for-purpose analytics and AI to authorized data, and SukariSol Connect, which links existing systems and devices through documented interfaces and controlled data exchange. The platform is designed to complement, not automatically replace, current systems. It will begin with stakeholder research and a demonstrator linking farm, collection, mill delivery and weighbridge/payment records. Satellite indicators and AI outputs will be treated as estimates requiring validation. Benefits will be tested against agreed pilot baselines; no unmeasured impact is claimed.

Keywords: sugarcane; AI; geospatial intelligence; interoperability; logistics; weighbridge; traceability; digital agriculture

1. The opportunity

Crop readiness affects harvest plans; harvest plans affect truck allocation and mill intake; verified delivery records inform farmer accounts; factory output and stock influence distribution. Kenya’s National Agriculture Production Report 2024 identifies operational challenges in the sugar industry [2]. SukariSol proposes a shared operating layer that turns authorized data into clearer workflows and reviewable decisions. It will first map existing systems and services—including Miwa Bora—and identify complementary gaps rather than assume that existing solutions are absent [3].

CONNECTED WORKFLOW

FARM	HARVEST	MILL	PAYMENT	MARKET
Map & observe	**Plan & dispatch**	**Receive & weigh**	**Reconcile**	**Plan & distribute**

The 2026 innovation call seeks practical concepts with a credible path to application [1]. SukariSol’s differentiation is its integration-first approach: connect authorized records across the value chain, then apply appropriate analytics to the linked workflow.

2. One platform: micro-apps, intelligence and connection

SukariSol is a modular platform, not a set of isolated apps. Modules share identity, permissions, audit history and agreed data definitions. The initial release prioritizes one end-to-end workflow; other modules are staged according to user need, data readiness and partner validation.

CaneYield / CaneVision

Farm registration, boundaries, acreage, road links, crop stage, harvest history and satellite vegetation indicators. Satellite proxies are not direct yield measurements.

CaneDoctor

Crop-stage reminders, agronomy guidance, pest/disease reporting and links to extension services; recommendations require local validation.

CaneFlow

Harvest requests, readiness, truck capacity, routes, dispatch, mill receiving slots and delivery events.

FairCane / CaneTrust

Match authorized weighbridge and delivery records to price rules, deductions, farmer statements and payment status; route discrepancies to review.

MillMind

Equipment registers, inspections, faults, repair history and downtime. Predictive maintenance only if reliable history supports it.

CaneCycle / ByProduct Hub

Discover eligible bagasse, molasses, filter cake and other materials, accounting for current uses, contracts, quality, handling, storage and transport.

SugarMarket

Use authorized sales, orders and stock records for demand scenarios, inventory visibility and distribution planning.

SugarShield

Trace approved farm, delivery, processing and distribution events, including quality/lot records where available.

ShambaScore / SukariSol Capital

Explore consent-based, explainable financing-readiness insights; never an automatic credit decision.

OutGrower OS

Grower registration, consent/contracts, field visits, input distribution, extension, harvest plans and organization reporting.

Miwa Mkononi

Farmer-facing mobile experience for alerts, farm details, delivery/payment status and support; low-bandwidth/offline-friendly capture where feasible.

Command Center

Role-specific dashboards, exceptions and action queues for mills, grower organizations, logistics and other authorized teams.

Two platform-wide capabilities

SUKARISOL INTELLIGENCE

AI and analytics for geospatial/field indicators, harvest readiness, route and truck planning, operational forecasts, and data-quality/anomaly flags. Use simple rules and statistics first; apply machine learning or computer vision only where suitable, validated data exists.

SUKARISOL CONNECT

A vendor-neutral integration layer using APIs, webhooks, scheduled CSV/Excel exchange, approved database views or device-specific connectors where supported. Includes field mapping, validation, timestamps, source tracking, access control, retries and audit logs.

3. How AI and integrations work in practice

Example: smart weighbridge connection

Where a mill’s weighbridge system permits integration, SukariSol Connect can import a ticket with a ticket identifier, truck/vehicle reference, timestamp, gross weight, tare weight and net weight. The platform can match the ticket to a collection order, farm or grower account and mill delivery record. SukariSol Intelligence can flag possible duplicate tickets, missing fields, unusual weight patterns, inconsistent timestamps or mismatches between related records. Flags are prompts for authorized staff to investigate—not proof of fraud, automatic accusations, or permission to alter an official weight. Verified outcomes can then feed FairCane/CaneTrust reconciliation with a traceable audit history.

1. CAPTURE	2. VALIDATE	3. MATCH	4. FLAG FOR REVIEW	5. RECONCILE

Other practical AI/analytics use cases

Geospatial & field view

Display farm boundaries and satellite observations; compare indicators over time; combine crop stage, field visits and harvest history. Cloud cover, resolution and ground truth constrain interpretation.

Harvest & logistics

Use transparent scheduling rules and road/travel estimates to propose collection windows and truck assignments. Compare predicted versus actual waiting and travel times before optimizing.

Farmer and transaction insights

Show authorized, explainable operational summaries such as delivery history, unresolved discrepancies or service needs. Use consent and human review; do not infer sensitive traits or make unsupported judgments about farmers.

Mill operations and markets

Summarize downtime and maintenance history, flag recurring events, and model stock/demand scenarios only when source records are adequate.

Interoperability by design

Connectors will be selected after inventorying the target system, vendor support, data formats, ownership, security and permitted use. Integration may be real-time or batch; the paper does not assume every device exposes an API. Data exchange will use documented mappings, validation rules, source timestamps, error handling and audit trails. Where direct connection is not available or approved, a controlled file import or clearly labelled simulation can demonstrate the workflow. Existing platforms remain systems of record where appropriate; SukariSol will not silently overwrite their records.

First demonstrator

Register/map farm → record crop stage and estimated harvest window → create collection order → assign truck → import or simulate weighbridge ticket → match delivery to farmer/account → reconcile against authorized payment records → show status and review flags to the right users. Distinguish working features, simulated data and planned features in every demonstration.

4. Implementation pathway

Month 1	Interview farmers, grower organizations, transporters, agronomists, mill operations/maintenance, finance and distributors; map workflow and pilot interest.
Month 2	Inventory farm, dispatch, weighbridge, payment, maintenance and stock data; assess rights, quality, device connectivity, satellite constraints and existing platforms.
Month 3	Prototype with users; select scope and integration method; agree permissions, baseline metrics and pilot evaluation.
Month 4	Build shared identity, farm map, collection/delivery workflow, a sample weighbridge connector/import and reconciliation view.
Month 5	Test access, calculations, audit logs, data quality, mobile usability and integration failures; prepare training and pilot report.

5. Evaluation, safeguards and adoption

Pilot measures should be agreed with the partner before deployment and compared with a baseline. No benefit is guaranteed in advance.

Area	Indicators to test
Farm intelligence	Boundary completeness; indicator agreement with field observations; forecast error only where reliable ground truth exists.
Logistics	Request-to-collection time, truck waiting time, empty travel, punctuality and mill queue time.
Weighbridge/reconciliation	Ticket-match rate, duplicate/mismatch rate, time to resolve exceptions, reconciliation completion and audit completeness.
Mill and market	Downtime-record completeness, maintenance response, forecast error, stock imbalance and traceability completeness.
Adoption	Active use, task completion, user comprehension, support requests and record completeness.

Architecture and safeguards

USERS & ORGANIZATIONS

Farmers • grower groups • drivers • mill teams • distributors

EXPERIENCES

Command Center • Miwa Mkononi • role-specific dashboards and alerts

MICRO-APPS

CaneYield/CaneVision • CaneDoctor • CaneFlow • FairCane • MillMind • CaneCycle • SugarMarket • SugarShield • ShambaScore • OutGrower OS

SHARED INTELLIGENCE + CONNECT

AI/analytics • APIs/files/device connectors • validation • identity • consent • permissions • audit logs

AUTHORIZED DATA SOURCES

GIS/satellite • farm/harvest • dispatch/delivery • weighbridge/payment • maintenance • inventory/orders • approved external services

Key safeguards: collect only necessary data; document ownership and permitted use; apply least-privilege access, consent where required and audit trails; protect personal and commercially sensitive records; label uncertainty; validate AI outputs; provide human review for consequential flags; and retain source records rather than silently changing them. Financing-readiness insights require consent, explainability and review before lender use. By-product listings must respect existing uses, contracts, quality requirements and applicable rules.

Adoption and sustainability

Initial pilot partners may include one willing mill or out-grower institution and a defined set of farms and transport workflows. Potential revenue models—enterprise subscription/licence, implementation and migration, training, integrations and validated advanced analytics—must be tested with customers. Scale only when the pilot demonstrates user value, trustworthy data, manageable support costs and a clear operating model. Use a modular monolith and geospatial-capable relational database initially; separate services only when justified.

Conclusion

SukariSol connects farm intelligence, movement of cane, mill receipt, farmer transactions and market planning through a governed platform. Its core distinction is the combination of fit-for-purpose AI with flexible, auditable integration of existing platforms and devices. It will complement existing services, validate data and demonstrate one measurable workflow before expanding. Success will be judged by adoption, trusted records and evidence from a partner-led pilot—not by AI claims alone.

References

[1] Kenya Sugar Board. “Call for Innovations 2026,” Sugar Industry Innovation Symposium & Expo. https://sugarinnovation.go.ke/call-for-papers/

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Declaration: SukariSol is a proposed concept, not a report of a completed system or measured pilot results. Projected benefits remain to be tested. Authors should confirm current call requirements and portal details before submission. The current call may require a telephone contact; add the authors’ verified phone number(s) if required.